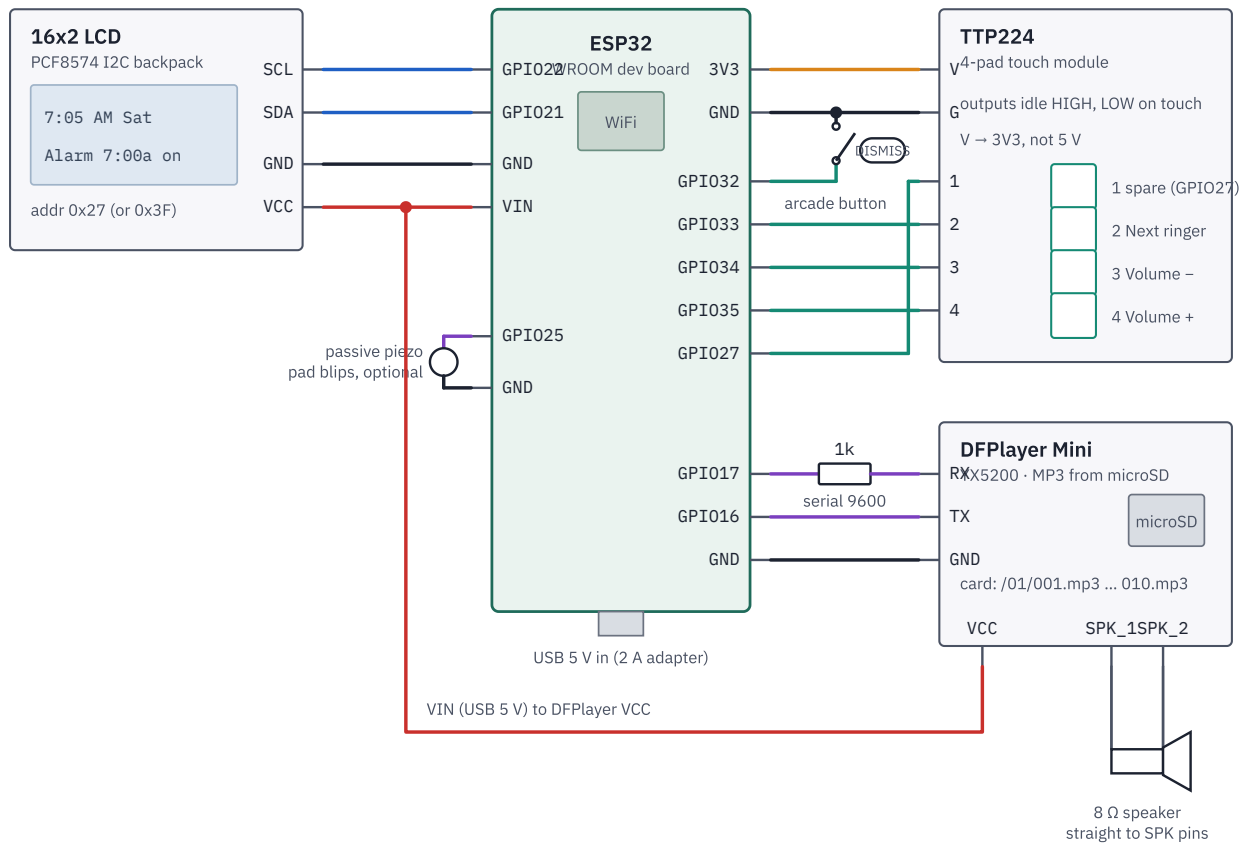


Kids Alarm Clock Wiring

ESP32 · 16x2 LCD (PCF8574) · arcade dismiss button · TTP224 touch pads ·
DFPlayer Mini MP3 · 5 V USB

Build per clock ×2
Rev 7 · 2026-09-17 · arcade = stop, pad 2 = next sound, pad 1 spare
kids-alarm-clock / docs



5 V (VIN) 3V3 GND I2C touch pads serial / piezo speaker

READ THIS FIRST

- Pin *positions* on the drawing are schematic. Find each pin by its printed label; GND appears several times on an ESP32 and any one is fine.
- LCD backpack runs on 5 V (VIN). At 3.3 V the contrast is too faint to read.
- Arcade button: microswitch **COM to GND, NO to GPIO32**, NC empty. Any GND pin. Internal pull-up handles it.
- TTP224 **V goes to 3V3** so its outputs stay at 3.3 V. GPIO34/35 are input-only pins, which is fine for this module. Pad 1 goes to GPIO27 (spare input, no job yet). Our modules idle HIGH and pull LOW on touch; firmware is set for that. If every pad reads ON untouched on a new module, flip `btn_inverted` in that board's YAML.
- DFPlayer runs on 5 V. Its TX at 3.3 V logic is safe into GPIO16. The **1k on the RX line** is required: without it the player hisses and can misread commands.
- Speaker goes straight to SPK_1 / SPK_2. The player has its own 3 W amp, so no resistor, diode or MOSFET module.
- Use a 2 A USB adapter. The player draws bursts near 1 A at full volume.
- Piezo is a passive buzzer between GPIO25 and GND, for pad-press blips only. Leave it off if you don't have one.
- Wait about 2 s after power before expecting the player to respond; the firmware already does.

Pins, parts, bench checks

Companion page to the wiring schematic

CONNECTIONS

WIRE	ESP32	GOES TO	ROLE
 red	VIN	LCD VCC · DFPlayer VCC	5 V from USB, one junction feeds both
 orange	3V3	TTP224 V	3.3 V so pad outputs are ESP-safe
 black	GND	LCD GND · TTP224 G · DFPlayer GND · piezo -	Common ground, four wires
 blue	GPIO21	LCD SDA	I2C data
 blue	GPIO22	LCD SCL	I2C clock
 teal	GPIO32	arcade button NO (COM → GND)	Dismiss (stops the alarm)
 teal	GPIO27	TTP224 1	Spare, wired for later (wakes backlight)
 teal	GPIO33	TTP224 2	Next ringer, 5 s preview (Snooze lives on the HA card)
 teal	GPIO34	TTP224 3	Volume down
 teal	GPIO35	TTP224 4	Volume up
 violet	GPIO17	1k → DFPlayer RX	ESP transmit, commands to the player
 violet	GPIO16	DFPlayer TX	ESP receive, "track finished" replies
 violet	GPIO25	piezo +	Pad blips (optional)
 speaker	—	SPK_1 → speaker → SPK_2	Either way round

Not used: GPIO 0, 2, 12, 15 (boot strapping pins). GPIO26 reserved for backlight dimming later.

PARTS PER CLOCK

- ☐ **ESP32 WROOM dev board**, 30 pin, USB-C + data cable
- ☐ **16x2 LCD** with PCF8574 I2C backpack
- ☐ **Arcade button**, Sanwa OBSF-24 (microswitch COM / NO / NC)
- ☐ **TTP224** 4-pad touch module (all four pads wired, pad 1 spare)
- ☐ **DFPlayer Mini**, YX5200 chip
- ☐ **microSD card**, 32 GB or less, FAT32, folder 01 with 001.mp3–010.mp3 (Track 1–10 in HA)
- ☐ **Speaker**, 32 mm full-range, 8 Ω 3 W
- ☐ **1 kΩ resistor** for the RX line
- ☐ **Passive piezo** (optional)
- ☐ **5 V USB wall adapter**, 2 A
- ☐ Dupont wires: 4 LCD, 2 button, 5 touch, 5 DFPlayer, 2 speaker, 2 piezo

BENCH CHECKS WITH BENCH.YAML

- ☐ **Power up** from a USB port. LCD backlight on, ESP LED on, DFPlayer LED off until it plays.
- ☐ **I2C scan** in the log shows the backpack. If it says 0x3F, change address in both YAMLS.
- ☐ **LCD** prints "Bench OK" and a counting uptime. Blank boxes only: turn the contrast pot on the backpack.
- ☐ **Arcade button** plays /01/001.mp3; the player's blue LED lights while playing. **Pad 2** stops it.
- ☐ **Pads 3 / 4** step volume down / up while a track plays.
- ☐ **Silent, LED never lights:** card not read. Reformat FAT32, check folder/file names, reseal the card.
- ☐ **Hiss or garbled commands:** the 1k on RX is missing, or TX/RX are swapped.
- ☐ **Log shows "DFPlayer finished track"** after a song ends. That event is what loops the alarm.
- ☐ **Then flash alarm-board-1** and press Test Alarm in HA.

DFPLAYER MINI PINS (16-PIN, CARD SLOT AT THE TOP)

LEFT COLUMN, TOP → BOTTOM	CONNECT	RIGHT COLUMN, TOP → BOTTOM	CONNECT
VCC	ESP VIN (5 V)	BUSY	open
RX	1k from ESP GPIO17	USB-	open
TX	ESP GPIO16	USB+	open
DAC_R	open	ADKEY2	open
DAC_L	open	ADKEY1	open
SPK_1	speaker	IO2	open
GND	ESP GND	GND	open (same net)
SPK_2	speaker	IO1	open

Everything you need is in the left column. Confirm against the silkscreen: some clones print the pin names on the underside.